PURE CUBIC THUE EQUATIONS ARE SOLVABLE IN DETERMINISTIC EXPONENTIAL TIME

THE DIOPHANTINE-FABLE EXPEDITION

Abstract. We prove that pure cubic Thue equations $x^3 - d\,y^3 = m$ ($d \geq 2$ cube-free, not a cube; $m \neq 0$; input size $s = O(\log d + \log |m|)$ in binary) are decided, with full solution lists, by a deterministic, unconditional algorithm in time $2^{O(s)}$. No worst-case bound of this shape appears in the literature: the best effective height bounds (Bugeaud–Győry) are exponential in the coefficient *bit-size*, so enumeration is doubly exponential, and the closest prior (Smart, ANTS-II 1996) is an explicitly practical per-method estimate, not a worst-case bound. The method walks the rank-one unit orbit of norm-m representatives from the Buchmann–Williams infrastructure and confines Thue-shape indices to a two-sided window: an elementary $O(1)$ forward bound, and a backward bound quasi-linear in the regulator R, via Sha's Matveev-based endgame re-run with the true root-ratio gap $3R/2$, against which the R in Matveev's height parameter cancels. A prototype matches PARI's certified `thue` on 1120/1120 grid instances; an independent reimplementation matches 13/13, including three high-index fields. All constants are explicit modulo one flagged final evaluation.

1. Introduction

1.1. **Smale's fifth problem and the Thue gap.** Smale's fifth problem (*Mathematical problems for the next century* [Sma98]; see also Lagarias's quotation of the AMS reprint in [Lag79], §1.4) asks for an algorithm deciding, given $f \in \mathbb{Z}[u, v]$, whether $f(x, y) = 0$ has a solution in $\mathbb{Z}^2$, in time 2^{s^c} for a universal constant c — exponential time $2^{\mathrm{poly}(s)}$ in the input size s. The size measure is *dense*: $s(f) = \sum_{|\alpha| \leq \deg f} \max(\log |a_\alpha|, 1)$, so every coefficient slot up to the total degree contributes at least 1. Smale notes that even decidability of the general problem is open.

Within this landscape, Thue equations $F(x, y) = m$ with F an irreducible binary form of degree ≥ 3 occupy a deceptively classical-looking stratum. Their finiteness is Thue's theorem; their effectivity is Baker's. Yet the best general effective height bounds — Bugeaud–Győry [BG96] — bound $\log \max(|x|, |y|)$ *polynomially in the coefficient magnitude* H, that is, **exponentially in the coefficient bit-size** $\log H$. Enumeration up to such

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Draft — authorship placeholder.

a bound therefore costs doubly exponential time in s, and the following gap results:

> Even Thue equations of **fixed degree 3** are not known to be decidable in time 2^{s^c} by the height-bound route.

This paper closes that gap for the pure cubic family. Throughout, $d \geq 2$ is cube-free and not a perfect cube, $m \neq 0$, and $s = O(\log d + \log |m|)$ denotes the bit-size of the input (d, m) in binary.

1.2. **Prior work: effectivity and practicality, but no worst-case bound.** The algorithmic Thue literature is rich but states no worst-case complexity theorem of the shape we prove.

- **Tzanakis–de Weger** [TdW89] set up the now-standard practical pipeline (Baker bound, LLL-based reduction, small-solutions search). Their paper states no complexity bound; the necessary algebraic data — fundamental units and the factorization of m in the field — are *explicitly assumed known.*
- **Bilu–Hanrot** [BH96] extend the pipeline to high degree (replacing the r-dimensional LLL reduction by continued-fraction reduction of two-term forms). Again: no complexity bound; the phrase is "solve in reasonable time ... provided the necessary algebraic number theory data is available".
- **Smart** [Sma96] is the closest prior and the work we position against. It is a complexity analysis of the Tzanakis–de Weger method at fixed degree n, measured in $\log |m| + \log L(F)$ ($L(F)$ the sum of the coefficient moduli), and it is explicitly an analysis of practical difficulty — "I know of no analysis as to how difficult it is in practice to actually find all the solutions" [Sma96, p. 363]. The solution space is divided into small solutions (direct search), medium solutions (continued-fraction reduction), and large solutions (eliminated by linear forms in logarithms), and the announced outcome is "an overall exponential complexity bound in terms of $\log L(F)$" [Sma96, p. 364] — exponential in the input size — with the practical failure modes stated explicitly: large m, large regulator or class number, near-coincident roots of $F(x, 1)$.[1] Smart's conclusion, quoted verbatim in Stroeker's review of the paper (Zbl 0896.11009): "So although solving Thue equations is now considered trivial we seem to be a long way off

[1] Verified against the openly available portion of the primary text: the publisher's two-page preview of [Sma96] (pp. 363–364) confirms the fixed-degree setting, the complexity measure $\log |m| + \log L(F)$, the three-phase division of the solution space, the failure modes, and the announced "overall exponential complexity bound in terms of $\log L(F)$"; the concluding quotation is reproduced verbatim in R. J. Stroeker's zbMATH review (Zbl 0896.11009). The remainder of the text (pp. 365–373) is paywalled; per-phase exponent shapes reported in secondary sources — e.g. a small-solutions cost of $O(|m|^{1/(n-2)})$ — are therefore not quoted above, and nothing in this section depends on them: the gap claim rests on the sweep of the books and surveys, not on the exact exponents of [Sma96].

from a 'good' algorithm". It is a per-method practical estimate — not a rigorous worst-case decision theorem, and no bound of the shape of Theorem 1 appears there.
- **PARI/GP** [PARI] prints one genuine unconditional complexity statement for `thue`: when $F(x, 1)$ has *no real root* the equation can be solved unconditionally in time $|m|^{1/\deg F}$. Every pure cubic has a real root, so even this rank-limited printed bound does not cover our family; and `thue`'s output is GRH-conditional unless certified.

A systematic literature verification across the books and surveys (Smart's 1998 book, Hanrot's LLL survey, Evertse–Győry, Waldschmidt, Gaál, Kim, Gherga–Siksek) found effectivity or practicality statements only. The theorem below therefore appears to be a citable gap in the literature, positioned — as we have just done — against [Sma96].

1.3. The main theorem.

Theorem 1 (Main Theorem)**.** *There is a deterministic, unconditional algorithm that, given cube-free $d \geq 2$ (not a cube) and $m \neq 0$ in binary, with input size $s = O(\log d + \log |m|)$, decides whether*

$$x^3 - d\,y^3 = m$$

has a solution in $\mathbb{Z}^2$ — and lists all solutions — in time $2^{O(s)}$.

The algorithm is fully specified and its correctness proof is complete; every constant in the analysis is effective and explicitly computable. The single item deferred — flagged wherever it occurs — is the *numeric evaluation* of one absolute constant chain (Remark 7.3): the theorem holds as stated, with the window bound of Lemma C semi-explicit, i.e. explicit modulo final constant evaluation.

1.4. **Proof strategy: seven steps.** Let $K = \mathbb{Q}(\alpha)$, $\alpha = \sqrt[3]{d}$, a complex cubic field with unit group $\langle -1, \varepsilon \rangle$ of rank one. A solution (x, y) is an element $\gamma = x - y\alpha$ of norm m; all norm-m elements fall into finitely many orbits under the unit action; and "being of Thue shape" is the vanishing of one coordinate — the α^2-coordinate — along the orbit, which is an integer linear recurrence of order 3 in the orbit index k. The algorithm (Section 3):

(1) compute the regulator, the Voronoi cycle, and the fundamental unit by the Buchmann–Williams rank-one infrastructure — deterministic, unconditional, no class group needed;
(2) enumerate the ideals of norm $|m|$ in O_K and extract a generator for each principal one, by the same cycle walk (Lemma A′);
(3) unit-reduce each generator (Lemma U);
(4) set up the order-3 integer recurrences (forward and reversed) for the cleared α^2-coordinate;
(5) bound the forward window elementarily: vanishing indices $k \leq 3$ (Proposition F);

(6) bound the backward window by one application of Matveev's theorem with the true root-ratio gap $3R/2$: vanishing indices $-k \le N_C = \widetilde{O}(R+s)$ (Lemma C);
(7) walk the windows with exact integer arithmetic, collect the zeros, recover (x, y), and apply the mandatory final filter $x^3 - dy^3 = m$.

The technical heart is Step 6, and it turns on the guiding principle of this program, stated here once and referred forward to Remark 4.6: **always ask exponential in what** — every bound must be calibrated against the *input* size s, never against an auxiliary quantity such as the regulator, for the regulator itself is exponential in the input, $R = 2^{\Theta(s)}$ (Section 2.3). A black-box application of Sha's explicit Skolem-type theorems [Sha19] gives a window of size $e^{22.5R}$ — which is $2^{O(s)}$ only under a *redefined* size measure containing R itself; calibrated against the input it is doubly exponential (Remark 4.6 preserves this near-miss as a cautionary lesson). The repair is not a new theorem of transcendence theory but a re-run of Sha's own endgame with the instance's true data: the two maximal-modulus roots of the reversed recurrence are the conjugate complex pair, the third root is smaller by the exact factor $e^{3R/2}$, and in the resulting inequality the R inside Matveev's height parameter $A_3 = 4R$ **cancels** against the gap $3R/2$ — leaving a window quasi-linear in R, hence $2^{O(s)}$ in honest input size.

1.5. **Deciding without exhibiting: the Pell analogy.** The philosophy is the one the Pell equation teaches one rung down. The fundamental solution of $x^2 - Dy^2 = 1$ has $\Theta(\sqrt{D})$ digits — exponential in the bit-size of D — yet solvability questions about generalized Pell equations are decided by walking orbits and comparing positions, never by expanding the fundamental solution unless it is actually needed. One rung up, the obstruction is steeper: by dense size $s = 20$ the fundamental unit of $\mathbb{Q}(\sqrt[3]{d})$ already reaches **1593 digits** per coefficient (Section 6.2, all records certified), against roughly 145 digits for the Pell envelope at the same size. The infrastructure machinery of Buchmann–Williams [BW88] keeps every *per-step* object at polynomial size — minima are represented by poly-size ideal data, "whereas the order of magnitude of a minimum can be as large as $\exp\sqrt{D}$" [BW88, Prop. 2.11] — and full expansion is invoked only where the $2^{O(s)}$ budget explicitly covers it. Stated precisely, the philosophy is that the algorithm never *enumerates solutions* — objects whose digit counts are exponential in s — not that it never writes down a large number. At the exponential-time target of this paper, *expanding* the fundamental unit and the Binet data is a legitimate simplification, adopted deliberately in Lemma A′: plain exact products along the cycle suffice. Compact representations become essential only for the stronger targets — poly-size certificates, NP membership — discussed in Remark 7.5.

1.6. **Organization.** Section 2 fixes notation and collects the field-theoretic preliminaries, including the two regulator floors the proof leans on. Section 3 states the algorithm. Section 4 proves the four supporting results (Lemmas B, U, A′, C and Proposition F), with all corrections from independent verification applied. Section 5 assembles the proof of Theorem 1. Section 6 reports the computational companions. Section 7 collects remarks and open problems.

2. Preliminaries

2.1. **The field, its embeddings, and its units.** Fix cube-free $d \geq 2$, not a cube, and write $d = ab^2$ with a, b coprime and squarefree. Let $\alpha = \sqrt[3]{d}$ (real) and $K = \mathbb{Q}(\alpha)$, a cubic field of signature $(1, 1)$: one real embedding σ_1 and one conjugate pair $\sigma_2, \sigma_3 = \bar{\sigma}_2$, with $\sigma_j(\alpha) = \omega^{j-1}\alpha$ for $\omega = e^{2\pi i/3}$. The discriminant is $-27a^2b^2$ if $d \not\equiv \pm 1 \pmod 9$ and $-3a^2b^2$ if $d \equiv \pm 1 \pmod 9$; in particular, for every pure cubic field

$$|\mathrm{disc}(K)| \geq 108,$$

the minimum being attained at $d = 2$. We will use this floor twice.

The unit group is $O_K^\times = \langle -1, \varepsilon \rangle$ of rank one. The torsion subgroup is $\{\pm 1\}$: any root of unity in K is fixed by the real embedding σ_1, hence equals ± 1. Normalize the fundamental unit by

$$\lambda := \sigma_1(\varepsilon) > 1, \qquad R := \log \lambda,$$

which is always possible (replace ε by $\pm\varepsilon^{\pm 1}$). Write $\mu = \sigma_2(\varepsilon)$; then $|\mu| = |\bar{\mu}| = \lambda^{-1/2}$, so the embedding moduli of ε are $(\lambda, \lambda^{-1/2}, \lambda^{-1/2})$ and R is the regulator of K.

Lemma 2.1 ($N(\varepsilon) = +1$ always)**.** *With the normalization $\lambda > 1$, the fundamental unit of a complex cubic field has norm $+1$.*

Proof. $N(\varepsilon) = \sigma_1(\varepsilon)\,|\sigma_2(\varepsilon)|^2 = \lambda|\mu|^2 > 0$, and a unit norm is ± 1. □

This one-liner is load-bearing: it makes the *reversed* orbit recurrence integral (Step 4).

2.2. **The index formula and the clearing constant** $3b$**.** By Dedekind [Ded00] (see [AW04] for a standard modern account), for $d = ab^2$ cube-free,

$$[\,O_K : \mathbb{Z}[\alpha]\,] \;=\; b \cdot \begin{cases} 3 & \text{if } d \equiv \pm 1 \pmod 9, \\ 1 & \text{otherwise.} \end{cases}$$

Consequently the index always divides $3b$, and for every $g \in O_K$, writing $g = c_0(g) + c_1(g)\alpha + c_2(g)\alpha^2$ with $c_i(g) \in \mathbb{Q}$, the cleared coordinates $3b\,c_i(g)$ are rational integers. All integer recurrences below are stated for the $3b$-cleared α^2-coordinate. We emphasize — because a draft of this work got it wrong, and empirical testing reveals that the naive "clear by 3" recipe fails at 91 values of $d < 500$ (e.g. $d = 12, 45, 175$) — that clearing by 3 alone is **not** sufficient: the correct clearing constant is $3b$.

2.3. **Regulator bounds, above and below. Above.** By Lenstra [Len92, Thm. 6.5], $hR \le |\Delta|^{1/2} \cdot \mathrm{polylog}|\Delta|$ at fixed degree; since $|\Delta| \le 27d^2 = 2^{O(s)}$, we get

$$R \;\le\; hR \;=\; 2^{O(s)}.$$

This is the honest calibration to keep in view throughout: the regulator is *exponential* in the input size, so any window or cost "polynomial in R" is exponential in s, and anything "exponential in R" is doubly exponential in s.

Below, twice. Astudillo–Díaz y Díaz–Friedman [ADF16, Thm. 7 and Table 1] prove the sharp lower bound for complex cubic fields:

$$R \;\ge\; 0.2811995743\ldots,$$

attained uniquely at the field of discriminant -23 (defined by $x^3 - x - 1$; the fundamental unit is the plastic number). Moreover, with exactly three exceptions — the fields of discriminant $-23, -31, -44$, with regulators $0.2811\ldots$, $0.3822\ldots$, $0.6093\ldots$ — every complex cubic field satisfies $R > 0.79$. Since pure cubic fields have $|\mathrm{disc}| \ge 108$, **none of the three exceptional fields is pure**, so:

$$\text{every pure cubic field satisfies } R > 0.79.$$

This floor is **load-bearing**: Matveev's theorem imposes the side condition $A_3 \ge |\log(\bar{\mu}/\mu)|$, and $|\log(\bar{\mu}/\mu)|$ can approach π. Our choice $A_3 = 4R$ satisfies the side condition precisely because $4R \ge 3.16 > \pi$ — a margin of 0.018. At the overall complex-cubic minimum $R = 0.2812$ the condition would *fail* ($4R = 1.12 < \pi$); the theorem survives on the pure-cubic discriminant floor by less than two hundredths. (The forward window, by contrast, needs only the unrestricted floor $R \ge 0.28119$.)

2.4. **Heights.** h denotes the absolute logarithmic Weil height. We use throughout:

- h is Galois-invariant: $h(\sigma_j(\gamma)) = h(\gamma)$.
- $|\log|\beta|| \le [\mathbb{Q}(\beta) : \mathbb{Q}] \cdot h(\beta)$ for $\beta \ne 0$.
- $h(\varepsilon) = R/3$: the Mahler measure of the minimal polynomial of ε is λ (the complex place, of local degree 2, contributes nothing since $|\mu| < 1$), so $h(\varepsilon) = \frac{1}{3}\log\lambda$. Note the complex place carries weight 2 in all such computations; in particular $h(\bar{\mu}/\mu) \le 2h(\varepsilon) = 2R/3$.
- $h(\omega) = 0$, $h(3d^{2/3}) = \log 3 + \frac{2}{3}\log d$.

2.5. **The coordinate extraction and the orbit recurrence.** For $g \in K$ the discrete-Fourier identity of the power basis reads

$$\sigma_1(g) + \omega\,\sigma_2(g) + \omega^2\sigma_3(g) \;=\; 3d^{2/3}\,c_2(g),$$

so the α^2-coordinate is a linear combination of the embeddings with weights of modulus $1/(3d^{2/3})$. An element $\gamma \ne 0$ and the orbit $g_k = \gamma\varepsilon^k$ give the

Binet form

$$c_2(\gamma\varepsilon^k) \;=\; b_1\lambda^k + b_2\mu^k + \bar{b}_2\bar{\mu}^k, \qquad b_1 = \frac{\sigma_1(\gamma)}{3d^{2/3}}, \quad b_2 = \frac{\omega\,\sigma_2(\gamma)}{3d^{2/3}},$$

and **all Binet coefficients are nonzero** whenever $\gamma \neq 0$ (the embeddings of a nonzero element are nonzero, and the DFT weights have modulus $1/(3d^{2/3}) \neq 0$). This verifies, once and for all, the implicit "all $b_j \neq 0$" hypothesis under which the Skolem-type machinery of Section 4 operates.

Since $[K:\mathbb{Q}] = 3$ is prime and $\varepsilon \notin \mathbb{Q}$, the characteristic polynomial of multiplication by ε is its (irreducible) minimal polynomial $X^3 - tX^2 + uX - 1$ (constant term $N(\varepsilon) = 1$ by Lemma 2.1), with simple roots $\lambda, \mu, \bar{\mu}$. Hence the cleared sequence

$$z_k \;:=\; 3b\,c_2(\gamma\varepsilon^k) \;\in\; \mathbb{Z}$$

satisfies the integer recurrence $z_{k+3} = t\,z_{k+2} - u\,z_{k+1} + z_k$ with *rational integer* coefficients — the parameter collapse that keeps every constant tame (in Sha's notation [Sha19], the Galois-closure degree of the coefficient field is 1). The reversed sequence $v_n := z_{-n}$ satisfies the recurrence with characteristic polynomial $X^3 - uX^2 + tX - 1$, the minimal polynomial of ε^{-1} — **integral because** $N(\varepsilon) = 1$ — with roots λ^{-1} (of modulus e^{-R}) and the conjugate pair $\mu^{-1}, \bar{\mu}^{-1}$ (of modulus $e^{R/2}$).

3. The algorithm

We display the algorithm, then discuss each step; the seven steps follow the verified skeleton exactly. Lemmas A′, B, U, C and Proposition F are stated and proved in Section 4.

3.1. Pseudocode.

```
Input:  cube-free d >= 2 (not a cube), m != 0, in binary;
        s = O(log d + log|m|).
Output: the complete list of (x, y) in Z^2 with x^3 - d y^3 = m.

Step 1  (Field data and infrastructure; no class group.)
        Factor d by trial division (cost O(d^{1/2}) poly = 2^{O(s)});
        write d = a b^2 with a, b coprime squarefree; extract the
        clearing constant 3b.
        Run BW88 Algorithm 2.13 on O_K: compute the Voronoi cycle of
        reduced principal ideals, the neighbor minima x_1, ..., x_p,
        the exact incremental products gamma_k = x_1 ... x_k, the
        fundamental unit eps = gamma_p, and the regulator R. [Lemma A']
        Expand eps in the integral basis (2^{O(s)} digits).

Step 2  (Norm-equation representatives.)
        Factor |m| by trial division. Enumerate all ideals I of O_K
        with N(I) = |m|  (at most d_3(|m|) of them; Dedekind basis,
        splitting at p | 3b read off from O_K/pO_K). For each I: test
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        principality by reducing I and searching the cycle stock; if
        principal, extract the generator
        gamma_I = mu_I * gamma_{k-1}^{-1} and verify (gamma_I) = I by
        one exact HNF check.                               [Lemma A']

Step 3  (Unit reduction.)
        Replace each generator gamma_I by gamma' = gamma_I * eps^j with
        | log|sigma_1(gamma')| - (1/3)log|m| | <= R/2.    [Lemma U]

Step 4  (Recurrence data.)
        For each gamma', form z_k = 3b * c_2(gamma' eps^k): an integer
        sequence satisfying the order-3 recurrence with characteristic
        polynomial minpoly(eps); the reversed sequence v_n = z_{-n}
        satisfies the integral reversed recurrence (N(eps) = 1).
        Expand eps and z_0, z_1, z_2 to their 2^{O(s)} digits.

Step 5  (Forward window.)   k_+ := 3.                 [Proposition F]

Step 6  (Backward window.)  N_C := c (R + log|m| + log d + 1) *
                                   log(R + log|m| + log d + 2). [Lemma C]

Step 7  (The walk and the filter.)
        For each representative gamma' and each sign sigma in {+1, -1}:
        iterate the exact integer recurrences over k in [-N_C, k_+];
        whenever z_k = 0, write sigma * gamma' eps^k = x - y alpha and
        KEEP (x, y) if and only if x^3 - d y^3 = m exactly
        (mandatory norm-sign filter).
        Output the collected set of solutions.
```

3.2. **Step 1 — regulator and unit, no class group.** The step opens by factoring d by trial division — cost $O(d^{1/2})\,\mathrm{poly}(s) = 2^{O(s)}$ — which yields the decomposition $d = ab^2$ and with it the clearing constant $3b$ consumed by Steps 2 and 4 (Section 2.2). At unit rank one the class group is never needed: principality of ideals is decided directly on the cycle of reduced principal ideals, so no certification burden attaches to Cl_K at all. We compute R and the cycle by Buchmann–Williams Algorithm 2.13 [BW88, Prop. 2.14]: deterministic $O(R\,D^{\epsilon})$ bit-operations ($D = |\Delta|$), unconditional, published with proof. By Section 2.3, $R = 2^{O(s)}$, so this is $2^{O(s)}$. The fundamental unit in *expanded* form costs $2^{O(s)}$ digits — affordable at this budget, and needed in Step 4 anyway. (Lenstra [Len92, Thm. 5.5] — whose deterministic exponent is $|\Delta|^{3/4}$, not $1/2$, and which Lenstra himself hedges with "appears to be true" — is a fallback citation only; the BW88 route is the one we rely on.)

3.3. **Step 2 — norm-equation representatives.** Trial-divide $|m|$ (cost $2^{s/2}\mathrm{poly}(s)$). The ideals of O_K of norm $|m|$ number at most $d_3(|m|) = 2^{O(s)}$ (three-fold divisor function; the counting argument is the proof line of [Len92, Thm. 6.5]). They are enumerated prime-by-prime: for $p \nmid 3b$, factor $x^3 - d \bmod p$; for $p \mid 3b$ (the primes dividing the index), read the splitting off the finite algebra O_K/pO_K in the Dedekind basis; assemble prime-power blocks by exponent vectors weighted by residue degrees, and multiply into Hermite normal form, poly(s) per ideal. Per candidate ideal, principality and a generator are obtained by the baby-step cycle walk — decision cost $O(R\,D^\epsilon)$ per candidate [BW88, Prop. 4.5] — with the tracked generator supplied by Lemma A′. (The $R^{1/2}$-rate giant-step variant [BW88, Thm. 4.7] is decision-only and is not used.)

3.4. **Step 3 — unit reduction.** Each generator is reduced along the one-dimensional log-unit lattice (Lemma U, sharpened form): the reduced representative γ' satisfies

$$\big|\log|\sigma_1(\gamma')| - \tfrac{1}{3}\log|m|\big| \le \tfrac{R}{2}, \qquad \text{hence} \qquad \big|\log|\sigma_2(\gamma')| - \tfrac{1}{3}\log|m|\big| \le \tfrac{R}{4} \text{ and } \left|\frac{\sigma_2(\gamma')}{\sigma_1(\gamma')}\right| \le e^{3R/4}.$$

The windows of Steps 5–6 apply to the **unit-reduced** representatives — this hypothesis is part of their statements.

3.5. **Step 4 — orbit and recurrence.** As in Section 2.5: z_k is the α^2-coordinate of $\gamma'(\pm\varepsilon)^k$, cleared by $3b$ (**not** by 3; Section 2.2). The recurrence has order 3, characteristic polynomial equal to the minimal polynomial of ε, simple roots, and all Binet coefficients nonzero since $\gamma' \neq 0$ (DFT weights of modulus $1/(3d^{2/3})$); the coefficients are rational integers, so Sha's Galois parameter collapses to 1. $N(\varepsilon) = \lambda|\mu|^2 > 0$ forces $N(\varepsilon) = +1$ (Lemma 2.1), which makes the reversed recurrence integral. Expanding ε and z_0, z_1, z_2 to their $2^{O(s)}$ digits is part of this step's stated cost.

3.6. **Steps 5 and 6 — the two windows.** The forward window is elementary and absolute: vanishing at $k > 0$ forces $\lambda^{3k/2} \le 2e^{3R/4}$, giving $k_+ \le 3$ (Proposition F — which is exactly the empirical maximum, Section 6.1). The backward window is Lemma C: $v_n \neq 0$ for all $n > N_C = c\,(R + \log|m| + \log d + 1)\log(R + \log|m| + \log d + 2)$, an absolute explicitly computable c — quasi-linear in R, hence $N_C = \widetilde{O}(R + s) = 2^{O(s)}$.

3.7. **Step 7 — the walk.** For every representative × sign × index $k \in [-N_C, k_+]$: one exact integer recurrence iteration ($2^{O(s)}$ digits each, $2^{O(s)}$ total), collect the hits $z_k = 0$, recover (x, y) from the 1- and α-coordinates, and apply the **mandatory final filter** $x^3 - dy^3 = m$: the walk also emits norm-$(-m)$ elements, and only the filter separates the two signs. (Since c_2 is linear, the $\pm$ sequences share their zero set; the sign branch matters at recovery, where $\gamma \mapsto -\gamma$ swaps the norms $\pm m$.)

Completeness is ideal-theoretic and library-free: any solution $\gamma = x - y\alpha \in \mathbb{Z}[\alpha] \subseteq O_K$ generates an ideal of norm $|m|$ appearing in Step 2's list; the unit orbits under $\langle -1, \varepsilon\rangle$ — the full unit group, torsion $\{\pm 1\}$ by the real embedding — cover all generators of each principal ideal; and $\gamma \mapsto -\gamma$ swaps $\pm m$, which the $\pm$-walk covers. The formal proof is Section 5.

4. The supporting lemmas

Throughout this section γ denotes a nonzero element of O_K with $N(\gamma) = \pm m$, and $b_1, b_2, \bar{b}_2$ the ($3b$-cleared) Binet coefficients of its orbit sequence (Section 2.5), all nonzero.

4.1. **Lemma B: root-of-unity exclusion.**

Lemma B. *The quotient $\bar{\mu}/\mu$ of the complex embeddings of ε is not a root of unity.*

Proof. Suppose $(\bar{\mu}/\mu)^t = 1$ for some $t \geq 1$. Then $\sigma_2(\varepsilon^t) = \sigma_3(\varepsilon^t) = \overline{\sigma_2(\varepsilon^t)}$, so $\sigma_2(\varepsilon^t)$ is real. (The contradiction target is "$\sigma_2(\varepsilon^t)$ real" — not "ε^t real": $\sigma_1(\varepsilon^t)$ is always real, so that statement would be empty.) Since $[K : \mathbb{Q}] = 3$ is prime, either $\varepsilon^t \in \mathbb{Q}$, in which case ε^t is a rational unit, $\varepsilon^t = \pm 1$, contradicting $|\sigma_1(\varepsilon^t)| = e^{tR} > 1$; or $\mathbb{Q}(\varepsilon^t) = K$, in which case the three conjugates of ε^t are pairwise distinct, contradicting $\sigma_2(\varepsilon^t) = \sigma_3(\varepsilon^t)$. $\square$

The same argument applied to $|\lambda/\mu| = e^{3R/2} \neq 1$ shows the full sequence is non-degenerate (no ratio of distinct roots is a root of unity), so its zero set is finite; but the proof below needs only the $\bar{\mu}/\mu$ case.

4.2. **Lemma U: unit reduction at rank one.**

Lemma U (rank-1 reduction). *Let K be a complex cubic field, ε normalized with $\lambda = \sigma_1(\varepsilon) > 1$, $R = \log \lambda$. For every $\gamma \in O_K$ with $N(\gamma) = m' \neq 0$ there is $j \in \mathbb{Z}$ such that $\gamma' = \gamma\varepsilon^j$ satisfies*

$$\left|\log|\sigma_1(\gamma')| - \tfrac{1}{3}\log|m'|\right| \;\leq\; \tfrac{R}{2}.$$

Consequently $\left|\log|\sigma_2(\gamma')| - \frac{1}{3}\log|m'|\right| \leq R/4$, $\;|\sigma_2(\gamma')/\sigma_1(\gamma')| \leq e^{3R/4}$, *and* $h(\gamma') \leq \frac{1}{3}\log|m'| + R + O(1)$.

Proof. $|\sigma_1(\gamma)|\,|\sigma_2(\gamma)|^2 = |m'|$, and for $t(\gamma) := \log|\sigma_1(\gamma)| - \frac{1}{3}\log|m'|$ we have $t(\gamma\varepsilon^j) = t(\gamma) + jR$; take $j = -\mathrm{round}(t(\gamma)/R)$. The consequences follow from $|\sigma_2|^2 = |m'|/|\sigma_1|$ and from bounding the house of γ'. $\square$

This is a one-dimensional reduction in the log-unit lattice — genuinely trivial as mathematics. Two computational remarks are part of the record: computing j rigorously requires $\sigma_1(\gamma)$ to polynomially many digits via interval arithmetic, cost $2^{O(s)}$ on $2^{O(s)}$-digit inputs; and a borderline rounding at half-integers only widens the forward window by 1 — a slack already accounted for in Proposition F.

4.3. Lemma A′: tracked generators along the Voronoi cycle.

Lemma A′ (generators along the cycle). *Fix the degree and unit rank one, and let p be the cycle length, so $p < 2R/\log\tau+1$ with τ the golden ratio (Williams: $\varepsilon_0 > \tau^{p/2}$). The neighbor minima $x_1, \dots, x_p$ of the reduced-ideal cycle each carry* poly(s)*-size data* [BW88, Prop. 2.11]*, and the incremental exact products $\gamma_k = x_1 \cdots x_k$ ($2^{O(s)}$ digits each) satisfy: the $(k{+}1)$-st reduced principal ideal is $I_{k+1} = (\gamma_k)^{-1}$, and $\varepsilon = \gamma_p$. Hence every principality witness and ε itself are computable, expanded, in deterministic total time $2^{O(s)}$, with exact HNF verification of each ideal identity.*

Proof (with the corrected orientation and certification). The Voronoi chain at rank one is $I_1 = O_K$ and $I_{k+1} = (1/x_k)\, I_k$, where x_k is the neighbor minimum of I_k, **computed exactly by BW88 Algorithm 2.13 / Williams** [Wil85]. Exactness of the neighbor step is what certifies completeness of the stock: floats used for steering alone could skip a minimum and still pass every HNF test, so the per-step computation must be the exact one. Telescoping, $\gamma_k = x_1 \cdots x_k$ is the $(k{+}1)$-st minimum and $I_{k+1} = (\gamma_k)^{-1}$; after a full cycle, $\gamma_p = \varepsilon$ exactly, under the $\sigma_1 > 0$, increasing-λ normalization. poly(s)-size of each x_k follows from [BW88, Prop. 2.4] ($|N(x)| \le \sqrt{D}\, N(\mathfrak{a})$), [BW88, Prop. 2.7(i)] (log-step $< \log\sqrt{D}$), and denominators $\le \sqrt{D}$. Digit growth of the exact products is additive, $O(p \cdot \mathrm{poly}(s)) = 2^{O(s)}$ per γ_k and $2^{O(s)}$ in total, including the per-step exact HNF chain identity $I_{k+1} \cdot (x_k) = I_k$.

Candidate-side recipe (Step 2's need): given an ideal I of norm $|m|$, reduce exactly to obtain the relative minimum μ_I (poly(s)-size); I is principal if and only if the reduction lands on the cycle stock at some position k, and then its generator is

$$g \;=\; \mu_I \cdot \gamma_{k-1}^{-1}$$

(exact division on $2^{O(s)}$-digit integers), certified by one terminal exact HNF check $(g) = I$. □

Three remarks on the constants and the design, verified against the primary sources.

1. **Cycle length.** [BW88, Cor. 2.8] gives $R/\log\sqrt{D} < p \le jR/c_1$ with $(j, c_1) = (7, \log 4)$ at degree 3 — constants confirmed against the Math. Comp. text and against Williams [Wil86], whose scope is *all* complex cubic fields and arbitrary reduced lattices (the paper also has the sharper $\theta_8 > 4$, and $\varepsilon_0 > \tau^{p/2}$, whence $p < 2R/\log\tau + 1 \approx 4.16R$ as used in the statement). A venue correction carried in our records: Williams' *Continued fractions and number-theoretic computations* is Rocky Mountain J. Math. 15 (1985) [Wil85], not Pacific J. Math.; the Pacific J. Math. paper is [Wil86].
2. **Why plain products.** An earlier plan routed height bounds through composition-then-reduce distance-slip estimates. The simplification

adopted here: at the $2^{O(s)}$ budget one simply takes the plain incremental exact products of the at most $7R/\log 4$ neighbor minima along the cycle. Compact representations (Thiel) are a luxury needed only for poly-size *certificates*, not for the EXP algorithm.

(3) **What Lemma A$'$ delivers to Step 1.** The full cycle walk computes $\varepsilon = \gamma_p$ and R along the way; Steps 1 and 2 share one infrastructure computation.

4.4. Proposition F: the forward window is $O(1)$.

Proposition F (forward window). *Let γ' be a unit-reduced representative (Lemma U) and z_k the $3b$-cleared α^2-coordinate sequence of $\pm\gamma'\varepsilon^k$. If $z_k = 0$ with $k > 0$, then*

$$\lambda^{3k/2} \;\le\; 2\left|\frac{\sigma_2(\gamma')}{\sigma_1(\gamma')}\right| \;\le\; 2e^{3R/4}, \qquad \textit{so} \qquad k \;\le\; \frac{1}{2} + \frac{2\log 2}{3R} \;\le\; 2$$

using the sharp complex-cubic floor $R \ge 0.2811995743$ [ADF16]. Allowing the one-step rounding slack of Lemma U:

$$k_+ \;\le\; 3,$$

which is exactly the maximum observed empirically (Section 6.1).

Proof. By the DFT form (Section 2.5), $z_k = 0$ means $b_1\lambda^k = -(b_2\mu^k + \bar{b}_2\bar{\mu}^k)$, whence $|\sigma_1(\gamma')|\lambda^k \le 2|\sigma_2(\gamma')|\lambda^{-k/2}$ (the weights share the modulus $1/(3d^{2/3})$, which cancels; the $3b$ clearing also cancels). Rearranged: $\lambda^{3k/2} \le 2|\sigma_2(\gamma')/\sigma_1(\gamma')| \le 2e^{3R/4}$ by Lemma U. Taking logarithms, $k \le \frac{1}{2} + \frac{2\log 2}{3R}$, and with $R \ge 0.28119$ the right side is ≤ 2.15, so $k \le 2$; the possible half-integer rounding in Lemma U widens the admissible window by at most one index. $\square$

Note the phenomenon the empirical data confirm: after unit reduction the R's cancel — the forward window is not merely poly(s), it is *bounded by an absolute constant.*

4.5. Lemma C: the backward window, quasi-linear in R.

Lemma C (backward window). *Let γ be a unit-reduced representative with $N(\gamma) = \pm m$, and let v_n be the $3b$-cleared α^2-coordinate sequence of $\gamma\varepsilon^{-n}$ ($n \ge 0$). There is an absolute, explicitly computable constant c such that $v_n \ne 0$ for all*

$$n \;>\; N_C \;:=\; c\big(R + \log|m| + \log d + 1\big)\,\log\big(R + \log|m| + \log d + 2\big).$$

Proof. Write $v_n = b_1\lambda^{-n} + b_2\mu^{-n} + \bar{b}_2\bar{\mu}^{-n}$ with all $b_j \ne 0$ ($\gamma \ne 0$; Section 2.5), and set $\mu^{-1} = e^{R/2}e^{i\theta}$, $b_2 = |b_2|e^{i\varphi}$. The two maximal-modulus roots of the reversed recurrence are the conjugate pair $\mu^{-1}, \bar{\mu}^{-1}$ (modulus $e^{R/2}$), the third root λ^{-1} is smaller by the exact factor $e^{3R/2}$, and the quotient $\bar{\mu}/\mu$ of the maximal pair is not a root of unity (Lemma B).

If $v_n = 0$, then $2|b_2|e^{nR/2}|\cos(\varphi + n\theta)| = |b_1|e^{-nR}$, so with $|\cos x| \geq \frac{2}{\pi}\operatorname{dist}(x, \frac{\pi}{2} + \pi\mathbb{Z})$:

$$(\dagger) \qquad \operatorname{dist}\Big(\varphi + n\theta,\ \tfrac{\pi}{2} + \pi\mathbb{Z}\Big) \ \leq\ \frac{\pi}{4}\frac{|b_1|}{|b_2|}\, e^{-3nR/2}.$$

The distance on the left is measured exactly by the linear form in logarithms (with the corrected conjugation and exact factor 2, both verified numerically on two fields):

$$\Lambda_n \ =\ a\,\log(-1)\ +\ \log\frac{b_2}{\bar{b}_2}\ +\ n\,\log\frac{\bar{\mu}}{\mu}, \qquad a \in \mathbb{Z} \text{ odd},\ |a| \leq n+2,$$

with

$$|\Lambda_n| \ =\ 2\operatorname{dist}\Big(\varphi + n\theta,\ \tfrac{\pi}{2} + \pi\mathbb{Z}\Big) \quad \text{exactly.}$$

The degenerate case is void. $\Lambda_n = 0$ if and only if $\cos(\varphi + n\theta) = 0$, i.e. the conjugate-pair term of v_n vanishes — but then $v_n = b_1\lambda^{-n} \neq 0$. So at any actual zero $v_n = 0$ we always have $\Lambda_n \neq 0$, and Matveev's theorem applies.

Apply Matveev [Mat00] (as restated in [Sha19, §2.4]) with $k = 3$ logarithms in the field $L = \mathbb{Q}(\sqrt[3]{d}, \omega)$ of degree

$$D_0 = 6.$$

L **is** the Galois closure of K: $\sqrt{\operatorname{disc}(K)}$ generates $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\omega)$ for *every* pure cubic field (the discriminant is -3 times a square in both Dedekind types), and L is stable under complex conjugation and contains -1, $b_2/\bar{b}_2$, and $\bar{\mu}/\mu$. The conclusion is

$$\log|\Lambda_n| \ >\ -\,C_0\,A_1A_2A_3\,\log\big(e(n+2)\big), \qquad C_0 = 2^{40}\,D_0^2\,\log(eD_0),$$

where the 2^{40}-versus-2^{38} slack absorbs the factor 2 in $|\Lambda_n| = 2\operatorname{dist}$, and:

- $A_1 = \pi$;
- $A_2 \ \leq\ D_0\,h(b_2/\bar{b}_2) + \pi \ \leq\ 12H_1 + \pi$, where (heights are Galois-invariant, and the $3b$-clearing adds at most $\log 3d$):

$$h(b_2) \ \leq\ h(\gamma) + \log 3 + \tfrac{2}{3}\log d + \log 3d \ =:\ H_1 \ =\ O\big(h(\gamma) + \log d + 1\big);$$

- $A_3 = \max\big(D_0\,h(\bar{\mu}/\mu),\ |\log(\bar{\mu}/\mu)|\big) \leq \max(4R, \pi) = 4R$, using $h(\bar{\mu}/\mu) \leq 2h(\varepsilon) = 2R/3$ ($h(\varepsilon) = R/3$; the complex place carries weight 2). Matveev's side condition $A_3 \geq |\log(\bar{\mu}/\mu)|$ — a quantity which can approach π — is met because pure cubic fields have $|\mathrm{disc}| \geq 108$, hence $R > 0.79$ (the exceptional fields of [ADF16] at discriminants $-23, -31, -44$ are not pure), so $4R \geq 3.16 > \pi$ — **margin** 0.018.

Combining Matveev's lower bound with (†) and $|\log(|b_1|/|b_2|)| \leq 12H_1$ (via $|\log|b|| \leq \deg(b) \cdot h(b)$ and $\deg \leq 6$):[2]

$$\frac{3R}{2}n \ \leq \ 12H_1 + \log\frac{\pi}{4} \ + \ C_0\,\pi\,(24H_1+\pi)\,(8R)\,\log\bigl(e(n+2)\bigr).$$

Divide by $3R/2$: **the R cancels in the Matveev term**, leaving

$$n \ \leq \ \frac{8H_1}{R} \ + \ c_2\,(H_1+1)\,\log\bigl(e(n+2)\bigr), \qquad c_2 \text{ absolute and explicit.}$$

Resolving n against $\log n$ (via $x \geq 2A\log A \Rightarrow x/\log x \geq A$; the term $8H_1/R$ is $O(H_1)$ by the regulator floor) and inserting the unit-reduction bound $H_1 = O(\log|m| + R + \log d)$ (Lemma U) gives N_C as stated. In particular

$$N_C \ = \ \widetilde{O}(R+s) \ = \ 2^{O(s)}$$

— quasi-linear in R, in place of the fatal $e^{22.5R}$ of the black-box route (Remark 4.6). □

Final write-up bookkeeping, carried forward visibly from the verified notes: the absolute constant c is traced through the chain $2^{40}\cdot 36\cdot\log(6e)\cdot\pi\cdot(12H_1+\pi)\cdot 4R$ (before the R-cancellation), with $D_0 = 6$ throughout, and the branch conventions for $\log(-1)$ absorbed into $|a| \leq n+2$ — all mechanical; the numeric evaluation is carried out in Remark 7.3.

4.6. **Remark: why the black-box window does not suffice (a preserved lesson).** An earlier draft of this work cited Sha's Theorem 1.2 [Sha19] as a black box, with "window $2^{O(s)}$". The claim was wrong in the way that matters most in this subject: the s in that statement was the extraction's *redefined* size $\max(R, h(\gamma), \log|m|, \log d) \supseteq R$, and $R = 2^{\Theta(s_{\text{input}})}$. In honest input size, the black-box window $e^{22.5R}$ is **doubly exponential** — a measured instance: $d = 9986$ has $R \approx 2605$ and a black-box window exceeding $10^{25{,}454}$ steps. The exponential loss is entirely an artifact of the worst-case Mahler-type root-separation quantity in Sha's theorems, which is $e^{-\Theta(R)}$ for our instances even though the *true* ratio gap is $3R/2$ — exponentially larger. Substituting the true gap into Sha's own endgame (a change of the proofs, not of the theorems — which is why Lemma C is stated as ours, with Sha's Lemma 2.8 and the Matveev application as extracted ingredients) is what produces the quasi-linear window. This was the third instance of the same trap in this program's history, and it has earned a name we record for the reader: **always ask exponential in what.**

[2]The displayed combined inequality retains the verified notes' doubled parameters $24H_1 + \pi$ and $8R$, a factor-4 safety slack over the minimal admissible Matveev choices $A_2 = 12H_1 + \pi$, $A_3 = 4R$ listed above; the slack is absorbed into the final constant evaluation flagged in Remark 7.3 and does not affect the shape of N_C.

5. Proof of the Main Theorem

Theorem 1. *There is a deterministic, unconditional algorithm that, given cube-free $d \geq 2$ (not a cube) and $m \neq 0$ in binary, with $s = O(\log d + \log |m|)$, decides whether $x^3 - dy^3 = m$ has a solution in $\mathbb{Z}^2$ — listing all solutions — in time $2^{O(s)}$.*

Proof. We show the algorithm of Section 3 is sound, complete, and runs in time $2^{O(s)}$.

Soundness. Every pair the algorithm outputs has passed the exact integer filter $x^3 - dy^3 = m$ in Step 7, so no false positives occur — in particular the norm-$(-m)$ elements that the walk necessarily emits are discarded there.

Completeness (ideal-theoretic). Let $(x, y) \in \mathbb{Z}^2$ satisfy $x^3 - dy^3 = m$, and set $\gamma = x - y\alpha \in \mathbb{Z}[\alpha] \subseteq O_K$, so $N(\gamma) = m$. The ideal (γ) has norm $|m|$ and therefore appears in Step 2's enumeration; it is principal, so Step 2 (Lemma A′) finds a generator γ_I with $(\gamma_I) = (\gamma)$, and Step 3 replaces it by the unit-reduced γ' in the same orbit. Two generators of the same ideal differ by a unit, and the unit group is $\langle -1, \varepsilon \rangle$ — the full unit group, since the torsion is $\{\pm 1\}$ by the real embedding — so

$$\gamma \;=\; \pm\,\gamma'\,\varepsilon^k \quad \text{for some } k \in \mathbb{Z}.$$

The α^2-coordinate of γ vanishes (Thue shape), so the cleared sequence of γ' has $z_k = 0$. Since γ' is unit-reduced, Proposition F bounds the positive side, $k \leq k_+ = 3$, and Lemma C bounds the negative side, $-k \leq N_C$; both windows apply precisely because they are stated for unit-reduced representatives. Hence $k \in [-N_C, k_+]$, the walk of Step 7 visits index k, detects $z_k = 0$, and recovers from $\pm\gamma'\varepsilon^k$ both candidate pairs $\pm(x, y)$; the filter retains exactly the one(s) of norm m — among them (x, y). Both norm signs are covered: if $N(\gamma') = -m$, the solution appears through the sign branch $\gamma = -\gamma'\varepsilon^k$, since $\gamma \mapsto -\gamma$ swaps $\pm m$ at odd degree. No step of this argument appeals to any external solver; completeness is purely ideal-theoretic.

Cost accounting. Write $D = |\Delta| \leq 27d^2 = 2^{O(s)}$ and recall $R \leq hR = 2^{O(s)}$ (Section 2.3).

- *Step 1:* trial-division factorization of d, $O(d^{1/2})\,\mathrm{poly}(s) = 2^{O(s)}$; then $O(R\,D^{\epsilon}) = 2^{O(s)}$ bit-operations [BW88, Alg. 2.13, Prop. 2.14]; expanding ε costs $2^{O(s)}$ digits (Lemma A′).
- *Step 2:* trial division $2^{s/2}\mathrm{poly}(s)$; at most $d_3(|m|) = 2^{O(s)}$ candidate ideals, each assembled in $\mathrm{poly}(s)$; per candidate one cycle search, $O(R\,D^{\epsilon}) = 2^{O(s)}$ [BW88, Prop. 4.5], with generator extraction and one exact HNF verification (Lemma A′).
- *Step 3:* interval arithmetic to polynomially many digits on $2^{O(s)}$-digit inputs: $2^{O(s)}$ (Lemma U).
- *Step 4:* expanding z_0, z_1, z_2 and the recurrence coefficients: $2^{O(s)}$ digits, stated cost of the step.
- *Steps 5–6:* window sizes $k_+ \leq 3$ and $N_C = \widetilde{O}(R + s) = 2^{O(s)}$.

- *Step 7:* at most $(N_C + k_+ + 1)$ iterations per representative and sign; each iteration is exact integer arithmetic on $2^{O(s)}$-digit numbers; total $2^{O(s)} \cdot 2^{O(s)} = 2^{O(s)}$.

The product of a $2^{O(s)}$ number of representatives with $2^{O(s)}$ work each is $2^{O(s)}$, completing the proof. □

We record the honest caveat once more: to *run* Step 6 one needs a numeric value for c; the constant chain of Lemma C supplies one mechanically, and its final evaluation is deferred (Remark 7.3). The theorem is unconditional and its proof complete; the window constant is semi-explicit — explicit modulo final constant evaluation.

6. Computational companion

Two experiments and one independent reimplementation accompany the proof. Neither is part of the logical argument; both shaped it.

6.1. **The orbit-walk prototype: 1120/1120.** `experiments/cubic_orbit_prototype.py` implements the orbit walk with PARI/GP [PARI] supplying the algebraic data: representatives from `bnfisintnorm` over `bnfinit(x^3 - d, 1)`, the $\pm\varepsilon^{\pm k}$ walk over $k \in [-60, 60]$, Thue-shape detection on the α^2-coordinate, and the exact final filter $x^3 - dy^3 = m$. The validation grid is

$$d \in \{2, 3, 5, 6, 7, 10, 11, 12, 13, 15, 17, 19, 20, 22\}, \qquad m \in [-40, 40] \setminus \{0\}$$

— 14 fields × 80 right-hand sides = **1120 cases** — and the reference is PARI's `thue` over `thueinit(x^3 - d, 1)`, whose flag-1 certification is unconditional. Result: **1120/1120 exact solution-set agreement**, and across all hits the maximum vanishing index was

$$\max |k| = 3$$

— exactly the bound $k_+ \leq 3$ that Proposition F later proved (the prototype's fixed cutoff $K_0 = 60$ was the honest gap between prototype and theorem; the two windows of Section 4 close it). We stress the boundary between the proven algorithm and this practical code: the cutoff $K_0 = 60$ operates under the strong empirical window conjecture of Remark 7.4, not under the proven bound of Lemma C; the proven window N_C — of order $10^{17} \cdot H_1 \log H_1$ steps once the constants of Remark 7.3 are inserted — is $2^{O(s)}$ in theory but is not what the prototype executes. For completeness: `bnfisintnorm` returns a complete system modulo units of *positive* norm, and the $\langle -1, \varepsilon \rangle$ walk covers a superset of those orbits.

6.2. **Unit growth: the wall being dodged.** `experiments/thue_unit_growth.py` charts why any method that must expand unit data pays exponentially: for the family $x^3 - dy^3 = 1$ it records d whenever the digit-size of the fundamental unit's coefficients sets a record (regulators from `bnfinit`, GRH-conditional; every record row re-certified unconditionally with `bnfcertify`). By dense size $s = 20$ (at $d = 1721$, $R \approx 3669.4$) the coefficients reach

1593 digits, all records certified — a steeper envelope than the Pell layer (roughly 145 digits at the same size), one rung down. The data are in `experiments/data/cubic_unit_records.csv`. This is precisely the obstruction the infrastructure route dodges: the algorithm expands ε only inside its stated $2^{O(s)}$ budget, and every per-step object it manipulates is polynomially sized (Lemma A′).

6.3. **An independent reimplementation: 13/13.** An independent **from-scratch mini-implementation** of the algorithm, written separately from the prototype, was validated against certified `thue`: **13/13 exact agreements**, including the high-index fields

$$d = 12,\ 45,\ 175 \qquad ([\,O_K : \mathbb{Z}[\alpha]\,] = 2,\ 3,\ 5),$$

precisely the fields at which the naive "clear by 3" recipe fails (Section 2.2) — so the validation deliberately covers the corrected code path. The same reimplementation verified the linear-form analysis of Lemma C numerically on two fields and validated the Binet integer sequence against a Thue-shape zero.

7. Remarks and open problems

7.1. **General complex cubic forms (rank 1).** The natural next target is the general irreducible cubic Thue equation of negative discriminant, $F(x, y) = m$: the associated cubic field is again complex, the unit rank is again 1, and the entire skeleton — cycle walk, unit reduction, two-sided window, exact walk — carries over. What grows is bookkeeping: the ring $\mathbb{Z}[\alpha]$ attached to a general form is no longer generated by a pure cube root, the index and clearing analysis of Section 2.2 must be redone for the form's ring, and the height accounting in H_1 acquires the form's coefficients. We see no conceptual obstruction, and state it as the immediate open problem: extend Theorem 1 to all complex cubic forms.

7.2. **Totally real cubics (rank 2) are the honest hard case.** For totally real cubic forms the unit rank is 2 and the picture changes qualitatively: the vanishing locus lives on a $\mathbb{Z}^2$-lattice of units, the walk has no canonical successor, and no dominant root separates a forward window — the problem takes on the flavor of S-unit equations rather than of a one-dimensional Skolem instance. We know of no route to a $2^{O(s)}$ bound there and explicitly defer it; nothing in this paper's method addresses it.

7.3. **The constant chain (evaluated).** The constants of Lemma C are now fully explicit; we record the evaluation. With $D_0 = 6$ (exact: $\mathbb{Q}(\sqrt[3]{d}, \omega)$ *is* the Galois closure, Section 4.5), the absolute constant c in N_C is traced through

$$2^{40} \cdot 36 \cdot \log(6e) \cdot \pi \cdot (12H_1 + \pi) \cdot 4R$$

— i.e. $C_0 = 2^{40} D_0^2 \log(eD_0) = 2^{40} \cdot 36\,(1 + \log 6) \approx 1.105 \times 10^{14}$ against $A_1 = \pi$, $A_2 = 12H_1 + \pi$, $A_3 = 4R$ — before the R-cancellation. Dividing the zero inequality $\frac{3R}{2}\,n \le 12H_1 + \log\frac{\pi}{4} + C_0\,\pi\,(12H_1+\pi)(4R)\log(e(n+2))$ by $3R/2$ cancels the R in the Matveev term and leaves

$$n \;\le\; \frac{8H_1}{R} + K\,(12H_1{+}\pi)\,\log\bigl(e(n{+}2)\bigr), \qquad K \;:=\; \tfrac{2}{3}\,C_0\,\pi{\cdot}4 \;=\; 2^{40}{\cdot}96\,\pi\,(1{+}\log 6) \;\approx\; 9.258{\times}10^{14}.$$

Resolving n against $\log n$ exactly as in the proof of Lemma C (via $x \ge 2A\log A \Rightarrow x/\log x \ge A$; the term $8H_1/R$ is $O(H_1)$ by the regulator floor) yields the explicit window

$$N_C \;\le\; \max\Bigl(\frac{16H_1}{R},\; 4K\,(12H_1+\pi)\,\log\bigl(2eK\,(12H_1+\pi)\bigr)\Bigr),$$

whose leading coefficient $48K \approx 4.44\times10^{16}$ multiplies $H_1\log(\cdot)$; under the factor-4 safety slack of the displayed combined inequality in the proof of Lemma C (parameters $24H_1{+}\pi$ and $8R$), K is replaced by $4K \approx 3.703{\times}10^{15}$ and the leading coefficient by $\approx 1.78\times10^{17}$. Either way the coefficient governing N_C is of order 10^{17}, and the numeric value of c is thereby fixed; the elementary constants of Lemma U and Proposition F and the branch conventions for $\log(-1)$ remain absorbed into $|a| \le n+2$. Sha remarks [Sha19, §2.4] that sharper three-logarithm bounds (Mignotte's kit) carry side conditions that need not hold, so no off-the-shelf improvement to the 2^{40} is available.

7.4. **Relation to Smale's fifth problem.** In the stratification of Smale's fifth problem that this program maintains, genus-0 components with at least three places at infinity — the stratum that classically reduces to Thue equations — form the first wall past the conic/Pell layer. Theorem 1 breaches that wall at its thinnest point: one fixed family, degree 3, pure. The strong empirical window conjecture — every Thue-shape index of a unit-reduced representative satisfies $|k| \le C(1+\log|m|/R)$ — remains open (max observed index: 3), as do the general-form extensions above, and, a fortiori, the uniform question of the full stratum: is $\{f :$ every component of genus $0\}$ decidable in time $2^{O(s)}$ uniformly in the degree? The order-3 Skolem instance at the heart of our walk is classically decidable (Mignotte–Shorey–Tijdeman [MST84], Vereshchagin [Ver85]); our contribution is not decidability but an instance-specific, fully explicit, two-sided window that is exponential in the *input*, not in the regulator. We have not pursued poly-size certificates; compact representations are exactly the extra ingredient a certificate version would need (Lemma A′), a further direction we leave open.

7.5. **Beyond exponential time: the named obstructions.** Deterministic *polynomial-time* decision for pure cubic Thue equations is not a rewrite of this paper away: it faces three independent open problems, which we name so that the reader can see exactly where the exponential lives.

(1) **The unit.** Polynomial-time computation of the regulator and the fundamental unit — even in compact representation — is equivalent to the principal ideal problem and is a major open problem

of computational number theory, classically believed hard: it underlies infrastructure-based cryptography, and polynomial time is known only quantumly, by Hallgren-type algorithms [Hal07]. Note that even negative-Pell *decision*, one degree down, is in NP $\cap$ coNP (Lagarias [Lag79]) with no classical polynomial algorithm known.

(2) **The zero test.** Certified zero-testing of the recurrence values against their $2^{O(s)}$-digit height bounds via modular evaluation needs exponentially many primes; per-prime testing yields only randomized decisions (coRP-style, as in straight-line-program zero-testing), whose derandomization is open.

(3) **The window.** The scan window is quasi-linear in $R = 2^{\Theta(s)}$ unless the strong window conjecture of Remark 7.4 — empirical maximum index 3, Section 6.1 — is proven.

The natural next target is therefore NP membership — a cubic analogue of Lagarias's Pell certificates [Lag79] — for which the strong window conjecture is the enabler.

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